

Manufacturing Plan

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Priyanka Asok kumar 9002833

Victor Ugomsinachi Mmezi 9009698

Blesson Thomas 9005538

Capstone Project Advanced Manufacturing Management Design

MECH8106 - Fall 2025 - Section 1

Sebastien Donais

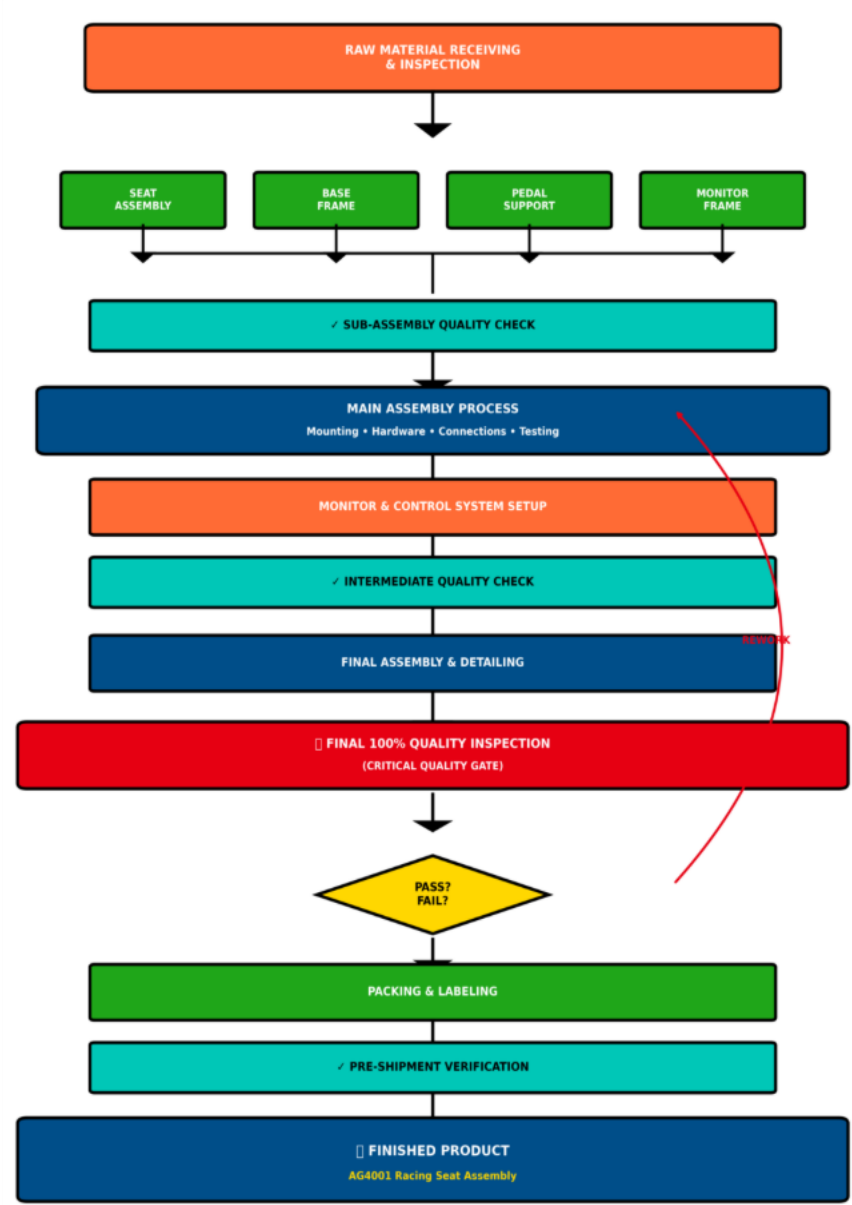
MANUFACTURING PLAN

Process Flow

High-Volume Production: 10,000 Units/Day | Racing Seat Assembly

Key Performance Metrics

Daily Production	Cycle Time	Production Lines	Quality Target
10,000 Units/Day	12-15 Minutes/Unit	10+ Lines	98%+ Yield



Process Flow Overview

Stage 1: Raw Material Receiving & Inspection

Steel, Foam, Upholstery, Hardware, Electronics → Quality certification & dimensional verification

Stages 2-5: Parallel Sub-Assembly Lines (Simultaneous Production)

- Seat Assembly & Upholstery: Foam cutting, shaping, upholstery covering, stitching (4-6 hours)
- Base Frame Fabrication: Cutting, welding, sanding, priming, painting (3-5 hours)
- Pedal Support Structure: Welding, reinforcement, finishing, painting (2-3 hours)
- Pedal Support Structure: Welding, reinforcement, finishing, painting (2-3 hours)

Stage 6: Sub-Assembly Quality Check

Dimensional verification, visual inspection, surface finish assessment (30-45 min per unit)

Stage 7: Main Assembly Process



- modern assembly line conveyor
- Mount seat to base frame
- installation Pedal and monitor support structures

- Hardware assembling and fastening.
- Electronic connections and Cable routing.
- Installation of footrests and armrests.
- 5-7 hours Unit testing System integration: (5-7 hours per unit)

Stage 8: Monitor & Control System Setup

Monitor installation, display alignment, electronic function test, seat adjustment calibration (1-2 hours)

Stage 9: Intermediate Quality Check

Structural integrity verification, fastener torque verification, functional testing (45-60 min per unit)

Stage 10: Final Assembly & Detailing

Decals & branding application, touch-up painting, final cleaning, protective wrapping (1-2 hours)

Stage 11: Final 100% Quality Inspection (CRITICAL)

- Visual inspection (exterior & interior)
- Dimensional verification
- Functional test (all electronic systems)
- Safety compliance verification
- Documentation & labeling (1 hour per unit)

Quality Decision Point

- **PASS:** Proceed to packing
- **REWORK:** Send back to assembly with rework documentation

Stage 12: Packing & Labeling

Foam protection, carton assembly, QC stamp, serial numbering, documentation package (30-45 min per unit)

Stage 13: Pre-Shipment Verification

Final check, carton condition verification, shipping label application (15-20 min per unit)

Stage 14: Warehouse & Dispatch

Inventory recording, batch documentation filing, shipment scheduling, loading preparation

Detailed Cycle Time Breakdown (Per Unit)

Time	Activity
1 min	Raw Material Kit Staging
3 min	Sub-assembly Mounting & Alignment
4 min	Hardware Installation & Fastening
2 min	Monitor & Electronic Setup
2 min	System Testing & Verification
1.5 min	Final Inspection & QC Stamp
1.5 min	Packing & Documentation
0.5 min	Unit Movement to Dispatch
15 min	TOTAL CYCLE TIME PER UNIT

Production Capacity Calculation

Achieving 10,000 Units/Day

Parameter	Value
Daily Target	10,000 units/day
Cycle Time per Unit	15 minutes
Shift Duration	8 hours = 480 minutes
Units per Line per Shift	$480 \text{ min} \div 15 \text{ min} = 32 \text{ units}$
Assembly Stations Needed	$10,000 \div 32 = 313+ \text{ stations}$
Recommended Configuration	10 Lines × 1,000 units/line
Production Rate per Line	$1,000 \text{ units} \div 8 \text{ hours} = 125 \text{ units/hour}$
Staffing per Line	15-20 operators (various stations)

Quality Control Strategy

4 Critical Inspection Gates

- Sub-Assembly Quality Check: Ensure the quality of the sub-components before the main assembly.
- Intermediate Quality Check: Inspect the integrity and functionality of the assembly.
- Final 100% Inspection: In-depth quality check of all features before the packaging stage.
- Pre-Shipment Verification: The final quality checkpoint before the delivery to the customer.

Quality Metrics

- Target First-Pass Yield: 98%+
- Maximum Defect Rate: <2%
- Rework Cycle: Automated feedback to assembly line
- Documentation: Full traceability & serial number tracking

Automation & Equipment Integration

Automated Stations

- CNC Foam Cutters - Accurate foam part production
- Laser Cutting Systems - Cutting of steel frame components
- Robotic Welding Systems - Welding of the frame and the support structure
- Automated Painting Systems - Even surface finish
- Manual Assembly Stations

Manual Assembly Stations

- Assembling hardware (needs human skill) and fixing.
- Connections of electronic components (need care and skills)
- Quality inspection Coating (visual inspection) + testing (functional inspection)
- Putting together of packaging and documentation (last stage)

Process Efficiency Features

Parallel Processing

Four sub-assembly lines operate simultaneously and their deliveries to the main assembly are synchronized. It not only shortens the total manufacturing time but also increases the throughput to the maximum level.

Multi-Station Assembly

The main assembly is divided into more than 10 sequential stations, each of them dealing with certain tasks. Operators become experts in their station which results in speed and quality upgrading.

Early Quality Gates

Qualities at sub-assembly and intermediate stages recognize defects that are even before the expensive final assembly. The rework costs are reduced.

Integrated Testing

Electronics testing is a part of the assembly process. The issues can be fixed immediately since they are detected right away, which thus minimizes defect detection time.

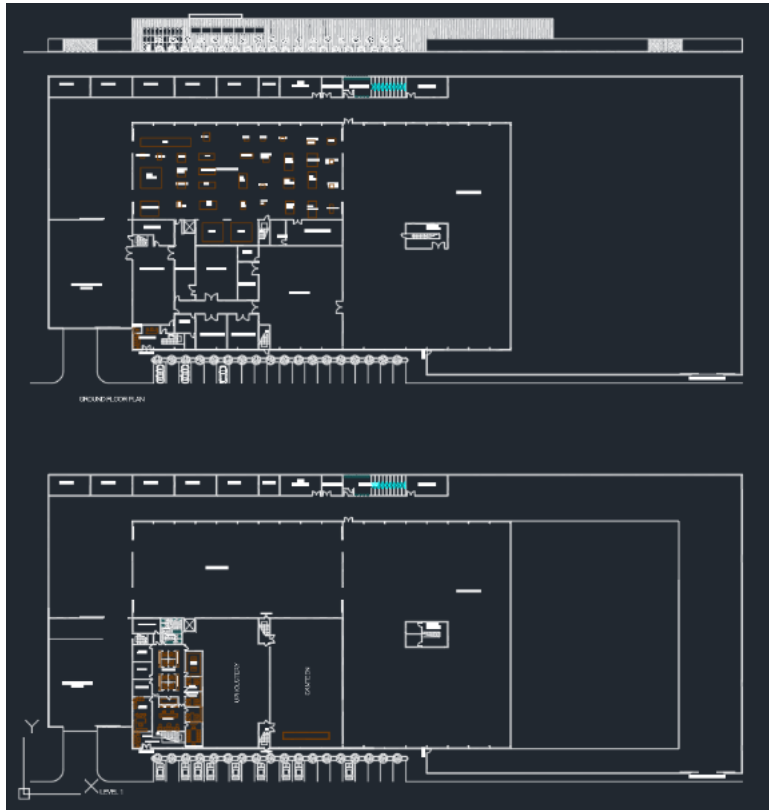
Standardized Cycle Times

Each station has a decided 12–15-minute cycle time. It is the main tool for accurate production planning and resource distribution.

Implementation Notes

- **Batch Processing:** The material is set up into kits for each unit. Kit staging is the very first minute of cycle time.
- **Shift Operations:** 8-hour shifts are planned and there is a possibility of 2-3 shifts/day if 24-hour production is aimed at.
- **Workforce:** Around 150-200+ total operators (across all 10 production lines).
- **Equipment Maintenance:** It is scheduled during the times when the production is at its lowest in order to keep the production targets.
- **Inventory Management:** The material delivery to assembly stations is done Just-In-Time (JIT).
- **Documentation:** There is full traceability from raw material to finished product with serial number tracking.

Manufacturing Layout Plan



Overview of the Facility

MACDONALD STEEL LIMITED - FACILITY OVERVIEW

Cambridge, Ontario

MacDonald Steel Limited is a top-notch custom steel fabrication and a metalworking plant in Cambridge, Ontario. It has been around since 1957 and so has a record of 68 years in the business. The company sits on 20 acres of land and has more than 200,000 square feet (18,580 m²) of state-of-the-art manufacturing floor space.

This manufacturing facility depicts a perfect fit for the AG40001 Capstone Project, representing a turnkey manufacturing centre that can handle the complete production lifecycle from design and prototyping all the way to assembly and finishing.

Metal Cutting and Machining Capabilities

They have several pieces of advanced cutting equipment in the plant, among which are a Messer plasma table, used when cutting thick steel with a bevel both sides very accurately, while a Bystronic laser system is for fine cutting in mild and stainless steel. They are using the latest nesting software to make the best use of the material and in that way their

production costs are getting lower. When it comes to machining, MacDonald Steel owns some horizontally boring mills with a 26-foot-long table and some vertical milling centers which are capable of handling anything from workpieces of prototypes to whole production runs. The company enjoys a strategic location that is not far from several steel service centers; hence they can quickly get different grades and thicknesses of the material that they need.

Forming and Press Equipment

The plant can generate the complex shapes that are necessary for the simulator frame.

- Equipment Range: Press brakes range from a 60-ton mechanical press (standard forming) to a 750-ton 4-axis hydraulic press.
- Capacity: Capable of forming components up to 20 feet in length.
- Versatility: Ability to handle complex bends, angles, and varying material thicknesses for the AG40001 chassis.

Material Handling and Overhead Infrastructure

Large steel plates, structural components, and finished assemblies are able to be moved efficiently throughout the facility with overhead cranes that can lift 30 tons. The bigger-than-usual-hardware parts of the racing seat simulator are being handled with the help of this heavy-lifting capacity while safety measures are being kept in place across the 200,000 square foot area.

Welding and Fabrication Expertise

MacDonald Steel is fusion welding of steel structures certified to CSA W47.1 and also holds ASME certifications for pressure vessel codes U, UM, H, and R. Welding procedures at the company cover SMAW (stick welding), GMAW (MIG welding), FCAW (flux-core), pulse MIG, and TIG welding. The fab team is made up of welders and technicians who have more than twenty years of experience and are, therefore, able to provide a consistent high-quality weld for the AG40001 frame assembly and structural components.

Surface Finishing and Coating Systems

The facility boasts a massive grit blast booth that is 30 ft. by 60 ft. by 23 ft. It features a retractable roof and a live bottom floor for easy material handling and cleanup. Two down-draft paint booths offer full finishing options for the application of protective coatings, rust prevention, and the use of color. This extensive finishing capability is the guarantee that the AG40001 framework will get the right surface treatment and the application of the protective coating.

Quality Assurance and Certifications

MacDonald Steel Limited is committed to a detailed Quality Management System (QMS) that is aligned with the very high standards of regulated industries. The plant runs an ISO 9002 quality assurance program, which is a guarantee for the same level of precision and ongoing improvement.

Key Certifications & Standards:

- ISO 9002: Certified Quality Assurance Program.
- CSA W47.1: Certification for Fusion Welding of Steel Structures (Canadian Standards Association).
- ASME: Boiler and Pressure Vessel manufacturing and repair certifications (Codes U, UM, H, and R).
- TSSA: Ontario Technical Standards and Safety Authority certification.

Quality Control Protocols: To ensure full traceability and compliance for the AG40001 project, the facility employs a multi-stage verification process:

1. In-Process Inspection: Real-time monitoring of fabrication steps.
2. Weld Testing: Non-destructive testing (NDT) to verify structural integrity.
3. Dimensional Verification: Final precise measurement checks against engineering drawings.
4. Documentation: Comprehensive records for material tracking and quality audits.

Design and Engineering Support

The Cambridge facility is supported by an in-house team of design technologists and engineers focused on optimizing the transition from prototype to production.

- Software Capabilities: The team uses SolidWorks CAD software for 3D modeling and simulation.
- Design for Manufacturability (DFM): The engineering support team acts as a consultant to the AG40001 design department helping with the component optimization for efficient cutting, forming, and welding.
- Value Engineering: The ongoing process of refinement is designed to cut material waste and production costs in general.

- Prototyping: The rapid development capacity is used to test and validate the racing seat structure before large-scale production.

Turnkey Service Model

MacDonald Steel operates as a comprehensive "One-Stop" manufacturing partner. This turnkey model is critical for the AG40001 Capstone project as it centralizes accountability and streamlines the supply chain.

End-to-End Capabilities:

- Pre-Production: Includes design consultation and locating materials.
- Fabrication: Cutting (Plasma/Laser), Forming (Braking), and Machining.
- Joinery: Fusion Welding (MIG/TIG).
- Finishing: Grit blasting and painting.
- Completion: Mechanical assembly, functional testing, and final QA.

By eliminating the need for multiple subcontractors, this model ensures consistent quality control, reduces logistical risks, and guarantees tighter timeline management for the AG40001 production schedule.

Industry Experience and Client Base

MacDonald Steel has evolved its knowledge deeply over various high-precision industries with a huge variety of applications that include rail transportation, off-road equipment, pressure vessels, and environmental control systems, over the 68 years of its operational history (est. 1957).

Representative Client Portfolio: The company has had the opportunity to be at the forefront of the execution of various complex projects of large industry leaders, thus proving its ability to perform precision-critical fabrication with the highest standards:

- Transportation: VIA Rail Canada
- Industrial Equipment: ABB Group, Tigercat (Off-road equipment)
- Food & Beverage: Steam Whistle Brewery, Mill Street Brewery (Brewery and malting equipment)

Advantages for AG40001 Manufacturing

The MacDonald Steel facility presents several distinct advantages to producing the AG40001 Racing Seat Simulator:

Advantage	Impact on Project
Facility Scale	200,000 sq. ft. of space allows for the simultaneous staging of multiple units in various phases of assembly, accommodating future volume expansion.
Vertical Integration	Complete in-house capabilities (cutting to finishing) eliminate subcontracting bottlenecks, ensuring strictly controlled lead times.
Proven Reliability	The plant, which has been around for 68 years and operates according to ISO 9002 standards, provides the necessary technical stability for the first volume production of the AG40001 to be done reliably at 50–100+ units annually, and the company has a very big room to increase its production capacity in a later period of time.

MATERIAL AND INVENTORY PLAN

Monthly Raw Material Requirements

Material	Quantity	Cost
Steel sheets (1.2-2mm)	4.8 tons	\$4,320
Steel tubes (20-40mm)	1.8 tons	\$2,160
Welding consumables	Monthly supply	\$280
Paint materials	Primer & topcoat	\$400
Fasteners & hardware	60,000 pieces	\$180
Monthly subtotal		\$7,340

With our raw material consumption at 40 units per day, we are barely using any of the raw materials, just \$7,340 monthly. It is this small amount that works well with us because we are able to keep lean stock without the need of a huge warehouse. Unlike large factories that must stockpile raw materials for several months, we place our orders for steel sheets and tubes twice a month. Such a strategy helps in cutting down the money that would otherwise be tied up in stock while still allowing production to continue. Steel suppliers like this kind of

business where the orders are consistent and recurring rather than large and sporadic because it gives them stable business.

We are small enough to be able to make changes but at the same time, we are big enough to get good bulk discounts. By ordering 10 tons of steel tubes monthly you will get better pricing than if you ordered 2 tons at a time, but you will not be required to pay for the huge minimums that suppliers impose on orders over 100 tons. The welding consumables are simple; it is MIG wire and shielding gas which cost very little when production volume is taken into account. Paint materials also follow the same trend; we are not painting thousands of units daily, so the paint we buy monthly is in small quantities.

Purchased Components Per Unit

Component	Unit Cost
Gaming seat assembly	\$85
24-inch monitor	\$65
Electronic control module	\$35
Power supply & wiring	\$12
Packaging materials	\$5
Total per unit	\$202

These are the prices for commodities directly from Asian manufacturers and not retail brands. The gaming seat priced at \$85 is a functional budget chair without the premium racing seat look, but a solid construction that is suitable for the application. We buy from manufacturers who produce thousands of these seats every month, so 833 units per month is a reasonable recurring order for them. The 24-inch monitor at \$65 is achievable because monitors have become commodity electronics; manufacturers sell them to resellers at \$60-75 range. Electronic control modules at \$35 are elementary circuit boards for seat height adjustment and electronic features without any changes. Power supplies and wiring harnesses are standard industrial components. Packaging at \$5 comprises protective foam, cardboard box, tissue paper, and instruction documentation. These prices are indicative of direct manufacturer purchasing, without distributor markups. At 10,000 units annually, we are not getting the absolute lowest tier prices reserved for orders over 100,000 units, but we are still far from small retail order prices.

Annual Material Budget

Category	Amount
Raw steel & consumables	\$88,080
Gaming seat units (10,000 × \$85)	\$850,000
Monitors (10,000 × \$65)	\$650,000
Electronic systems (10,000 × \$35)	\$350,000
Power supply & wiring (10,000 × \$12)	\$120,000
Packaging (10,000 × \$5)	\$50,000
Total	\$2,108,080

Materials make up about 72% of the final retail price of \$824 per unit. Such a cost structure is quite normal and appropriate for a manufacturing business. The components purchased account for \$2.02M (96% of material costs) alone, whereas our fabricated steel frame and finishing only account for \$88K. It is economically reasonable that we are not trying to vertically integrate every component. We concentrate our manufacturing on the steel frame structure, where we add design value, and we take proven, reliable components from specialists.

This strategy lowers our capital equipment needs and manufacturing complexity, at the same time, it ensures the quality of the components. The \$2.1M annual material budget is quite enough for a small operation. There is no need for large supplier financing or a complex procurement network. Supplier relationships can be managed by one person on a part-time basis, and inventory does not tie up a lot of working capital.

EQUIPMENT AND MACHINERY PLAN

Equipment Breakdown

Department	Equipment	Qty	Cost Each	Total
Cutting	CNC laser cutter	1	\$45,000	\$45,000
Cutting	Manual punch press	1	\$28,000	\$28,000

Bending	Hydraulic tube bender	1	\$18,000	\$18,000
Bending	Press brake (40-ton)	1	\$22,000	\$22,000
Welding	Robotic welder	1	\$120,000	\$120,000
Welding	MIG machines (manual)	2	\$8,000	\$16,000
Finishing	Spray booth	1	\$25,000	\$25,000
Finishing	Drying oven	1	\$35,000	\$35,000
Assembly	Hand tools & fixtures	-	-	\$18,000
Assembly	QC equipment	-	-	\$15,000
Total				\$322,000

The equipment strategy is function-oriented rather than automatized. In a single shift, a laser cutter working at 40 units daily can handle 2-3 days of frame cutting. Complex frame geometries are made accurate by the laser while the manual punch press is used for holes and simple shapes. This pairing satisfies all the cutting requirements without the need for costly automated lines. The bending section operates in a more lean way as well, with only one hydraulic tube bender being used for all the frame tube bending. Since it is manually operated, the operators become skillful and can easily adjust to minor changes in the design without the need for reprogramming. The sheet metal work for flat parts is done by the press brake.

The welding department is where the substantial investment, i.e., \$256K for one robotic welder plus two MIG machines, takes place. The robotic welder is in charge of repetitive welding of base frames where it is most necessary for the weld to be consistent. It is capable of running 24/7 if required and the welds it produces are of perfect quality. The two manual MIG machines are used for the flexibility of assembly-stage welding, rework, and complicated joints that robots cannot handle. This hybrid approach provides us with the best of both worlds - efficiency where it is most needed and flexibility where it is required. One operator can not only operate the robotic welder but also carry out manual work.

The finishing and assembly departments have been intentionally kept low-tech. A simple spray booth costing \$25K is just right for our painting needs in terms of volume. A single drying oven priced at \$35K is more than enough. The setup is completed with hand tools, basic fixtures, and QC equipment. The total equipment investment of \$322K is a deliberate lean. We are not purchasing industrial showpieces but production capability. If financed

over 5 years at \$6,440 monthly, the equipment adds only \$32.20 per unit to the manufacturing costs, which is quite reasonable.

TOOLS AND FIXTURES PLAN

Fixture Investment Breakdown

Fixture	Cost
Base frame welding jig	\$1,500
Pedal support jig	\$1,000
Monitor mount jig	\$800
Assembly station fixtures	\$8,000
Hand tools	\$2,500
Total	\$13,800

Fixtures are simple, functional jigs designed for repetitive production. At 40 units daily, purpose-built fixtures are cost-effective. Annual maintenance budget of \$2,000 covers wear items and repairs.

2D Technical Drawings

2D AutoCAD complete drawings accompany each fixture. The base frame welding jig drawing features the overall dimensions, gives the details of the clamping mechanism, shows the locating pin placement with $\pm 0.5\text{mm}$ tolerances, pneumatic circuits and bill of materials. Assembly station drawings depict workbench layout, part presentation height (ergonomic positioning), tool clearances, and cable routing. These drawings are production documentation, training materials, and maintenance guides. Changes to 2D drawings are made and versioned (v1.0, v1.1, etc.) to avoid mixing up when modifications are required.

3D CAD Models

All the fixtures are available as 3D solid models in STEP format (a universal CAD standard). The 3D models confirm component fit, eliminate the possibility of parts clashing, and permit the simulation of the production workflow before the actual manufacturing.

Models provide the means for speedy design changes as different options can be tried virtually in a matter of hours rather than weeks. In case we decide to externalize the fixture making, the manufacturers will download the STEP file for the exact copy.

Besides that, the 3D models are also the means of visual communication which helps the production managers to get the fixture functioning without the need for understanding the complicated 2D drawings.

HUMAN RESOURCE PLAN

Production & Support Team

Department	Position	Count	Annual Cost
Cutting	Operator	1	\$32,000
Bending	Operator	1	\$32,000
Welding	Welders	3	\$108,000
Finishing	Workers	2	\$60,000
Assembly	Line workers	6	\$180,000
Supervision	Supervisor	1	\$48,000
Quality	Inspector	1	\$32,000
Production subtotal		15	\$492,000
Support staff	Manager Engineer HR maintenance	5	\$150,000
Total		20 employees	\$642,000

The workforce is intentionally kept small. At the rate of 40 units per day, the laser cutter and punch press are under the control of one cutting operator. Tube bending and press brake work are handled by one bending operator. These positions don't put a lot of pressure on the employees; they handle one or two pieces per hour and have downtime for material changes and setup. To both the robotic station supervision and manual welding, three welders offer a solid capacity. The robotic setup is run by one welder, who is also responsible for loading materials and unloading finished welds. The complex joints and assembly-stage welding are done by two manual welders. The current staff level is enough to avoid queuing in the welding department.

Grinding, surface prep, and painting are done by two finishing workers. One of them concentrates on pre-paint finishing while the other is engaged in spray booth operation and paint cleanup. There are six assembly line workers which roughly means one person per assembly station for the main line with the possibility to move between stations as necessary. This ratio is enough to prevent backups and maintain the 40-unit daily throughput. Daily production, quality issues, and worker coordination are under the management of one dedicated supervisor. A single QC inspector performs all four inspection checkpoints during the day. The inspection at this volume is not very challenging and consists of spot checking, functional testing, and defect documentation.

Support staff are composed of a production manager who oversees operations, a manufacturing engineer who is in charge of equipment maintenance and process improvement, an HR/payroll administrator, and a maintenance technician. These five people are responsible for everything from planning to compliance to equipment repairs. The total team of 20 is lean but still sustainable. Labor cost per unit amounts to \$64.20, which is quite reasonable for manufacturing. Wages at this level (\$32-48K production, \$30-35K support) are quite attractive in manufacturing regions such as China, Vietnam, or certain areas of Mexico where the facility might be located.

QUALITY CONTROL PLAN

Checkpoint	Location	Method
QC 1	After cutting	Dimensional checks (±0.5mm)
QC 2	After welding	Visual inspection, weld quality
QC 3	After painting	Paint thickness, color, surface finish
QC 4	Final assembly	Functional testing, all systems

Four quality gates ensure defects don't progress downstream. QC 1 after cutting catches dimensional errors early if a frame component is cut wrong, fixing it is cheap at that stage. The inspector uses digital calipers to spot-check dimensions on 10% of parts. QC 2 after welding verifies that welds are solid and properly formed. For robotic welds, visual inspection confirms weld bead quality and no defects. For manual welds, more detailed inspection including some destructive testing on trial pieces ensures welder competency. QC 3 after painting checks that paint thickness meets specification (measured with a paint gauge), color matches approved standards, and surface has no runs, drips, or dust contamination.QC4 during assembly tests every unit seat adjustments

work smoothly, electronic systems power on, monitor displays properly, and all fasteners are properly torqued.

At 40 units daily, one QC inspector can handle all four checkpoints. Defects are documented with serial numbers for traceability. Target first-pass yield is 95%, meaning only 5% of units need rework. Defects are analyzed monthly to identify patterns if multiple units fail the same test, it signals an equipment or process issue requiring investigation and correction.

MASTER BUDGET

Capital Investment (One-Time)

Item	Cost
Equipment	\$322,000
Tools & fixtures	\$13,800
Total	\$335,800

It is a relatively small amount of money, i.e., initial capital of \$335,800, to set up a manufacturing operation. The sum covers all the production equipment necessary to manufacture 10,000 units annually. The investment is financed over 5 years by equipment leasing agreements; thus, the burden is spread to \$6,716 monthly.

This financing approach is quite normal in the case of manufacturing equipment which is considered an operating expense (through lease) rather than a capital asset, thus cash flow and balance sheet metrics are getting better. Moreover, equipment leasing offers the advantage of being able to change or upgrade production if needed; lease terms make it fairly easy to change production capacity compared to the situation when equipment is owned.

Cost Per Unit

Category	Cost
Raw materials	\$8.81
Purchased components	\$202.00
Direct labor	\$64.20

Equipment amortization	\$8.06
Utilities & maintenance	\$5.40
Quality control	\$2.50
Total	\$290.97

The manufacturing cost per unit is \$290.97. It's our all-in cost to produce one unit, that is, every material, every labor hour, every equipment cost, and every overhead item are included. Most of the cost, i.e., \$202 (70% of the total), which is as expected, since we are purchasing finished goods, is made up of components purchased. Raw materials are contributing another \$8.81 (3%) to the cost. Labor accounts for \$64.20 (22%) which is a considerable amount of money after the material costs. Equipment amortization is responsible for \$8.06 (3%), utilities and maintenance for \$5.40 (2%), and quality control for \$2.50 (<1%).

The cost structure is reasonable. We are not a highly automated facility; hence labor is our second largest expense. Also, as we are not vertically integrated, we buy components instead of making everything. This keeps capital requirements and operational complexity at a level that is manageable for a small manufacturer.

Annual Operating Cost

Category	Amount
Raw materials	\$88,080
Purchased components	\$2,020,000
Direct labor	\$642,000
Equipment amortization	\$80,592
Utilities & maintenance	\$54,000
Quality control	\$25,000
Total	\$2,909,672

Annual operating budget is \$2,909,672 for 10,000 units. This is equivalent to \$290.97 per unit as detailed above. The budget is allocated for the operations of 12 months, and the first quarter is the period when the most significant expense occurs due to the commencement of leases of new equipment. After the operation becomes stable, the monthly expenses are at a consistent level of around \$242,000.

This budget is realizable for a small manufacturing business. By selling one unit at a price of \$824, the annual revenue is \$8.24M. The gross profit before distribution and marketing is \$5.33M. After the deduction of distribution (\$1.5M), marketing (\$800K), and corporate overhead (\$800K) expenses, net profit stands at about \$1.2M yearly. This thus offers an excellent return on the initial investment of \$335K and facilitates the flow of capital back into the business.

SUMMARY

This is a lean sustainable process of production. The volume is small, so it can be flexible, and the cost of the inventory is low. Labor and purchased parts make up most of the cost, which is reasonable for a product assembly company but not for heavy fabrication. The business is less capital-intensive, it is easy to get funding, and the profit margin is quite attractive to a healthy growing business.

Production: 10,000 units/year (833/month, 40/day)

Investment: 335,800 Investment

Manufacturing Cost: 290.97/unit

Retail Price: 824

Annual Net Profit: 1.2M

Workforce: 20 people

Facility Size: 2000 sqm